

STACK

Q2)

Question:

DS	Session	Stack	Question Information	Level 1	Challenge 42
Question description					
Rajinikanth organized a technical round interview in an Animation company for the set of computer science candidates.					
the task is to implement stack operations for two stacks and merge the stacks into one.					
Get two sets of stack elements and reverse them, then merge them into one stack.					
Rajinikanth has given the deadline of only 15 minutes to complete the problem.					
Can you Help the candidates to complete the problem within the specified time limit?					
Function Description					
a) push(): Adds the new item at the beginning of the linked list using the first pointer.					
b) pop(): Removes an item from the beginning using the first pointer.					
c) merge(): Links the first pointer second stack as next of the last pointer of the first list.					
Constraints					
$0 < n, m < N$					
$1 < arr[i] < 1000$					
Input Format:					
First-line indicates n & m, where n is the number of elements to be pushed into the stack and m is the number of pop operations that need to be performed					
The next line indicates the n number stack elements					
Output Format:					
First-line indicates the top of the element of the stack					

PROGRAM:

```
test.c > main()
2
3  #define MAX 1000
4
5  int stack1[MAX], stack2[MAX];
6  int top1 = -1, top2 = -1;
7
8  void push1(int x) {
9      stack1[++top1] = x;
10 }
11
12 void push2(int x) {
13     stack2[++top2] = x;
14 }
15
16 int pop1() {
17     return stack1[top1--];
18 }
19
20 int pop2() {
21     return stack2[top2--];
22 }
23
24 int main() {
25     int n, m, x, i;
26
27     scanf("%d %d", &n, &m);
28
29     for (i = 0; i < n; i++) {
30         scanf("%d", &x);
31         push1(x);
32     }
33
34     for (i = 0; i < m; i++) {
35         scanf("%d", &x);
36         push2(x);
37     }
38
39     while (top1 >= 0) {
40         printf("%d ", pop1());
41     }
42
43     while (top2 >= 0) {
44         printf("%d ", pop2());
45     }
46
47     return 0;
48 }
```

OUTPUT:

TEST CASE 1:

```
5 8
9 8 7 6 5
5 6 7 8 3 11 2 3
5 6 7 8 9 3 2 11 3 8 7 6 5
```

TEST CASE 2:

```
10 4
91 18 17 16 15 15 16 17 18 13
11 12 13 14
13 18 17 16 15 15 16 17 18 91 14 13 12 11
```